

KSA Office IT Support & Infrastructure Management

Phase 07: IT Support Operations

Afaq Business Solutions Co.

Application Troubleshooting & Log Analysis Standard Operating Procedure (SOP)

1. Purpose

Provide a standardized process for diagnosing application errors through log review, configuration verification, safe recovery, and validation.

2. Scope

Applies to IT Support personnel investigating application incidents in controlled environments.

3. Application Incident Workflow

- Receive and document the incident.
- Review relevant application logs.
- Search logs for errors and warnings.
- Review surrounding log context.
- Identify the likely cause.
- Verify related application configuration.
- Apply authorized corrective action.
- Validate the configuration and application behavior.
- Document the outcome and close or escalate the ticket.

4. Log Analysis

- Review startup, processing, error, and termination events.
- Use targeted searches to isolate ERROR entries.
- Review surrounding entries to establish event sequence.
- Record relevant evidence without exposing sensitive information.

5. Configuration Verification

- Confirm the expected configuration file exists.
- Review required configuration values.
- Verify file accessibility and content.
- Avoid unrelated configuration changes.

6. Safe Recovery

- Use a controlled environment for recovery testing.
- Restore only the required configuration.
- Use approved configuration values.
- Record every corrective action.

7. Validation

- Confirm the recovered configuration exists.
- Verify required parameters.
- Test the affected application function when possible.
- Confirm the reported issue is resolved.

8. Evidence Handling

- Preserve relevant logs and investigation notes.
- Do not expose credentials, tokens, personal information, or confidential configuration.
- Store evidence in the designated Evidence directory.

9. Escalation

- Escalate when the issue cannot be safely corrected.
- Escalate suspected application defects or infrastructure failures.

10. Best Practices

- Diagnose before changing configuration.
- Make the smallest authorized change necessary.
- Validate after corrective action.
- Document findings and communicate the result to the user.